

Spatial Steering for Image Descriptions

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On my honor as a University Student, I have neither given nor received unauthorized aid on this assignment as defined by the Honor Guidelines for Thesis-Related Assignments

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Abstract

Image descriptions often emphasize color, size, or texture more than layout. This is especially important for instructional images, where users may need spatial relationships to understand how objects, steps, or parts of a scene are arranged. This paper studies whether activation steering can be used to identify and partially steer a spatial-versus-descriptive direction in caption-conditioned generation. I adapt the probability-weighted concept-vector method of Cyberek et al. [1], originally used for gender bias control, to steer image-description prompts toward spatial language. The method uses COCO-derived captions to construct prompts, measures each prompt’s next-token spatial-descriptive bias, learns a steering vector from the corresponding activation differences, and evaluates the impact of steering on generated continuations. In continuation-level evaluation, moderate steering makes generated continuations more likely to include object-location and scene-layout language while avoiding obvious degeneration in the tested setting. However, the useful steering range is limited because stronger steering can degrade output quality. Overall, the experiments find that spatial language in caption-conditioned generation is measurable and partly steerable at inference time, but the effect depends on continuation prefix, model choice, steering coefficient, and output-quality constraints.

1 Introduction

Image-description systems often describe what objects look like before they say where those objects are. They may mention color, size, material, or texture while underspecifying layout. For accessibility, that matters because spatial language can communicate scene structure: a bench in front of a gate, a bridge across a river, or a suitcase beside a person.

Prior work on accessible image descriptions shows that blind and low-vision users do not want one fixed style of description for every task [9, 10]. What makes a description useful depends on the user’s goal. When the goal is to understand layout, object relationships, or how a scene is arranged, spatial language can be especially important. Recent studies confirm this gap is real: blind and low-vision users specifically need spatially grounded guidance rather than purely visual descriptions [12], and current AI-generated descriptions score low on user satisfaction, suggesting the problem remains unsolved [11].

Activation steering is an inference-time method that shifts a model’s behavior by modifying its internal representations during generation, without changing any model weights [5, 6, 7]. In decoder-only transformers, each layer passes a hidden state vector to the next through the residual stream, and steering works by nudging that vector in a chosen direction [3, 4]. Following Cyberek et al. [1], this paper uses a two-stage pipeline: first, it learns a direction from contrastive examples, and then applies that learned direction to the residual stream during generation. Here, the analogous target is a visual-spatial concept rather than a gender concept. The goal is to learn a direction in the model’s residual stream whose projections correspond to spatial versus appearance-focused associations in caption continuation.

This paper tests that idea in a deliberately narrow text-only caption-continuation setting: the model receives an existing COCO caption as text, not the original image. The method first builds spatial and descriptive caption pools from COCO-derived captions, then uses those pools to learn a steering direction from residual-stream activations. That direction is selected using held-out validation prompts and then applied during generation to test whether continuations can be shifted

toward spatial language. The experiments evaluate both the strength of the learned direction and the tradeoff between stronger steering and output quality.

The rest of the paper is organized as follows. Section 2 describes the word sets, caption filtering process, vector extraction method, and evaluation metrics. Section 3 reports the standardized Qwen model comparison, the local next-token diagnostic, continuation-prefix comparisons, and qualitative examples. Sections 4 and 5 discuss what the results show, where the method is limited, and what future user-centered evaluation would require.

2 Method

This section describes the full steering pipeline. First, I define the spatial and descriptive word sets used to measure the target axis (subsection 2.1). Second, I use those word sets to filter COCO-derived captions into spatial and descriptive pools, convert captions into prompts, and learn a candidate steering vector from residual-stream activations (subsection 2.2). Third, I apply the selected steering vector during generation by modifying the model’s hidden state at the chosen layer (subsection 2.3). Finally, I define the normalized spatial ratio used to score generated continuations, while treating next-token disparity as a diagnostic for filtering and layer selection (subsection 2.4).

This pipeline follows Cyberek et al. [1], which learns and applies steering vectors for model-internal concept representations. The target concept is different here: instead of studying an internal gender representation, this paper studies a spatial–descriptive language axis in caption-conditioned generation. Spatial and descriptive word sets provide the measurable terms for this concept, but the intervention itself is applied to residual-stream activations rather than directly editing generated words.

2.1 Building the Spatial and Descriptive Word Sets

The first step is defining what counts as spatial or descriptive language. The lists were built from published lexical resources, LLM-assisted expansion, and manual filtering. Stanford’s Empath lexicon was used as a starting point for appearance-related categories such as color, size, and visual appearance [13]. The Visual Spatial Reasoning dataset was used as a starting point for spatial-relation language [14]. These seed terms were then expanded with Gemini 3.1 Pro [15] and manually reviewed to remove vague, duplicate, overly broad, or multi-word terms that do not correspond to single tokens in the model’s tokenizer.

After this review, the final lists contain 44 spatial terms covering position, direction, and layout (e.g., *left*, *behind*, *foreground*, *horizon*) and 60 descriptive terms covering color, size, shape, and condition (e.g., *red*, *large*, *round*, *broken*). The full lists are given in Appendix A. The final sets were reviewed to keep the spatial–descriptive axis focused on relation-and-layout language versus appearance language, rather than becoming a broad automatic classifier.

In the local pipeline, these terms are used only for filtering prompts, selecting layers, and scoring generated continuations. The model still produces probabilities over the full vocabulary during generation; the spatial and descriptive word lists are used afterward to measure how much spatial or descriptive language appears.

2.2 Learning the Steering Vector

This phase builds a spatial–descriptive direction from COCO-derived captions [8]. COCO contains human-written captions for everyday images with common objects and scene layouts, making it a practical source for studying spatial and appearance-based description language. It has three steps: filtering captions into contrastive pools, recording residual-stream activations, and selecting the best steering layer.

Caption filtering. For each COCO caption x , let $n_s(x)$ and $n_d(x)$ be the number of terms from the spatial and descriptive word lists. A lexical score

$$c(x) = n_s(x) - n_d(x) \quad (1)$$

assigns each caption to the spatial pool if $c(x) \geq 5$ or if $n_s(x) \geq 2$ and $n_d(x) = 0$, and to the descriptive pool if $c(x) \leq -5$ or if $n_d(x) \geq 2$ and $n_s(x) = 0$. Captions that do not meet either condition are dropped. The filtering process begins with 616,767 COCO caption examples and produces cleaner contrastive pools while excluding borderline captions. To keep the experiment computationally manageable and class-balanced, the training set is capped at 1,500 captions per class. The strongest-scoring captions are selected from each filtered pool, and the final split is then shuffled. Because both filtered pools contained at least 1,500 eligible captions, the final training set contains 3,000 captions total: 1,500 spatial and 1,500 descriptive.

Prompt filtering. After captions are turned into prompts, the model is used to score each prompt according to its next-token spatial–descriptive bias. Let T^+ be the set of tokens corresponding to spatial word-list terms and T^- be the set of tokens corresponding to descriptive word-list terms. For each prompt x_i , the model produces a next-token probability distribution $p_\theta(t \mid x_i)$ over the full vocabulary. The spatial and descriptive probability masses are

$$p^+(x_i) = \sum_{t \in T^+} p_\theta(t \mid x_i), \quad p^-(x_i) = \sum_{t \in T^-} p_\theta(t \mid x_i). \quad (2)$$

The prompt-level disparity score is then

$$s_i = p^+(x_i) - p^-(x_i). \quad (3)$$

A positive score means the model assigns more next-token probability mass to tokens from T^+ , while a negative score means it assigns more mass to tokens from T^- . Prompts are partitioned using threshold $\delta = 0.1$:

$$f(x_i) = \begin{cases} D^+ & s_i > \delta \\ D^- & s_i < -\delta \\ D^0 & |s_i| \leq \delta \end{cases} \quad (4)$$

Here, D^+ is the spatial-biased prompt pool, D^- is the descriptive-biased prompt pool, and D^0 is the low-disparity prompt pool. The D^0 prompts are called low-disparity rather than semantically neutral because they are defined only by the model’s next-token probabilities over the spatial and descriptive token sets. They are not the same as captions dropped by the earlier lexical filter, and

they may still contain some spatial or descriptive content. In this experiment, prompts in D^+ and D^- are used to learn the steering vector, while prompts in D^0 are used to estimate the activation offset for centering.

Vector extraction. After the prompts are sorted into spatial-biased, descriptive-biased, and low-disparity pools, the next step is to extract a candidate steering vector for each layer. For each layer ℓ , the residual activation h_x^ℓ is collected for each prompt x . Following the method of Cyberek et al. [1], the mean activation of the low-disparity pool, $\bar{h}_0^\ell$, is subtracted from each activation. This centers the activations before comparing the spatial-biased and descriptive-biased prompt pools.

For each pool, I then compute a weighted average activation:

$$\bar{h}_+^\ell = \sum_{x \in D^+} \tilde{w}_x (h_x^\ell - \bar{h}_0^\ell), \quad \bar{h}_-^\ell = \sum_{x \in D^-} \tilde{w}_x (h_x^\ell - \bar{h}_0^\ell) \quad (5)$$

Here, $\bar{h}_+^\ell$ represents the average centered activation for spatial-biased prompts, while $\bar{h}_-^\ell$ represents the average centered activation for descriptive-biased prompts. The weights are nonnegative and are normalized within each pool. For prompts in D^+ , weights are proportional to s_x ; for prompts in D^- , weights are proportional to $-s_x$. Equivalently, each pool uses weights proportional to $|s_x|$, so prompts with stronger next-token spatial or descriptive signals have more influence.

The unnormalized direction for layer ℓ is the difference between the normalized spatial-biased and descriptive-biased activation means:

$$u_\ell = \frac{\bar{h}_+^\ell}{\|\bar{h}_+^\ell\|} - \frac{\bar{h}_-^\ell}{\|\bar{h}_-^\ell\|}. \quad (6)$$

The steering vector used for intervention is then normalized:

$$v_\ell = \frac{u_\ell}{\|u_\ell\|}. \quad (7)$$

This produces one normalized candidate steering vector per layer.

Layer selection. Each layer produces a candidate steering vector, so the final step in vector learning is selecting which layer to use during generation. Each candidate vector is evaluated on held-out validation prompts. The validation activations are projected onto the candidate vector, producing one projection score per prompt. These projection scores are then compared with the prompt-level next-token disparity scores.

The goal is to choose a layer where the candidate vector best aligns with the spatial–descriptive distinction measured in the validation prompts. Following Cyberek et al. [1], the final 5% of layers closest to the output are excluded. For the local Qwen/Qwen-1.8B-chat run, layers 11–13 show the strongest validation signal, and layer 11 is used for the main steering experiments.

2.3 Applying the Steering Vector

At inference time, the selected steering vector is applied using the projection-based intervention from Cyberek et al. [1]. For each generated token, the hidden state at the selected layer is modified as follows:

$$h'_\ell = h_\ell - \text{proj}_{v_\ell}(h_\ell) + \lambda v_\ell \quad (8)$$

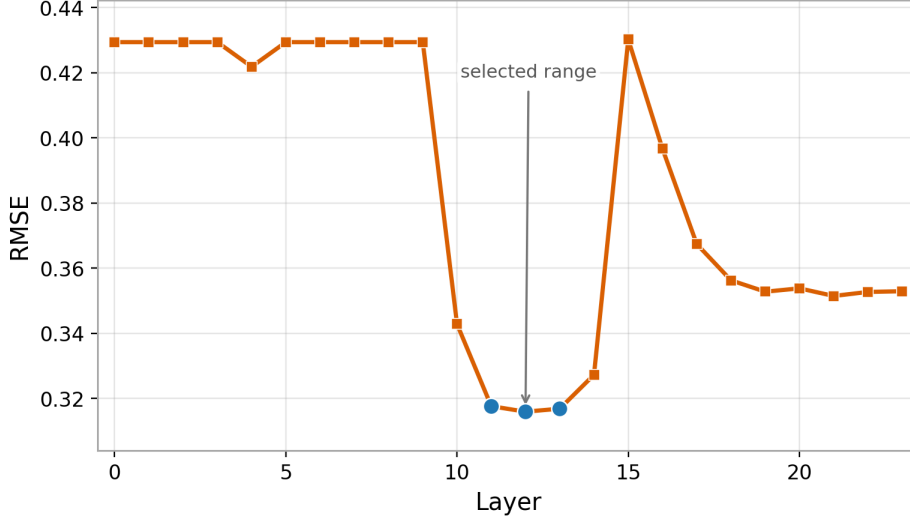


Figure 1: Layer-selection diagnostic for the local Qwen/Qwen-1_8B-chat run. Candidate layers are ranked by validation RMSE between the layer-wise projection diagnostic and held-out next-token spatial-descriptive disparity scores. Layers 11–13 have the lowest validation error, and layer 11 is selected for the main steering experiments.

Here, h_ℓ is the original hidden state at layer ℓ , v_ℓ is the normalized steering vector, and λ controls how strongly the model is steered.

This equation has two parts. First, the projection term removes the part of the hidden state that already points along the spatial-descriptive direction. Second, the λv_ℓ term adds back a controlled amount of that direction. When $\lambda = 0$, the intervention removes the hidden-state component along the learned axis without adding a positive or negative offset. Positive λ values push the model toward spatial language, while negative values push it toward descriptive language. This intervention only changes the hidden state during generation; the model weights are never updated.

Different continuation prefixes begin with different amounts of spatial or descriptive bias, so one λ value is not enough to describe the full behavior. For this reason, the results report coefficient-response curves across several λ values instead of focusing only on one best setting.

2.4 Normalized Spatial Ratio Metric

The prompt-level next-token disparity score s_i , defined above, is useful for filtering prompts and selecting a steering layer. To evaluate generated text, however, I use a continuation-level metric computed from the generated continuation itself.

For any generated text y , let $n_s(y)$ be the count of spatial word-list terms in y , and let $n_d(y)$ be the count of descriptive word-list terms in y . I define the normalized spatial ratio $R(y)$ as

$$R(y) = \begin{cases} \frac{n_s(y) - n_d(y)}{n_s(y) + n_d(y)} & \text{if } n_s(y) + n_d(y) > 0 \\ 0 & \text{if } n_s(y) + n_d(y) = 0. \end{cases} \quad (9)$$

This score ranges from -1 to $+1$. A score near -1 means the text contains mostly descriptive

word-list terms, while a score near +1 means it contains mostly spatial word-list terms. If the text contains no words from either list, it receives a score of 0 to avoid division by zero.

For a set of generated continuations $Y = \{y_1, \dots, y_N\}$, I define the mean normalized spatial ratio as

$$\bar{R}(Y) = \frac{1}{N} \sum_{i=1}^N R(y_i). \quad (10)$$

When the continuation set Y is clear from context, I write this mean score as $\bar{R}$. Positive values of $\bar{R}(Y)$ mean that the generated continuations contain more spatial word-list terms on average, while negative values mean they contain more descriptive word-list terms on average. Because $\bar{R}(Y)$ is computed from complete generated continuations, it measures the text the model actually produces rather than only the next-token probabilities.

3 Experiments and Results

This section evaluates the steering procedure in two stages. First, it tests whether the same procedure produces measurable continuation-level shifts across multiple Qwen models when the prompt format is held fixed. Second, it uses the local `Qwen/Qwen-1.8B-chat` run to examine how changing the continuation prefix affects steering behavior. This ordering presents the standardized model comparison first, then the more detailed local analysis of prompt sensitivity and qualitative behavior.

3.1 Experimental Setup

All experiments use a text-only caption-continuation setup. The model is given a COCO-derived caption, not the original image. The default prompt template is:

```
Describe this image: [COCO caption]
The image shows
```

The continuation prefix is appended before generation, so the model generates the remaining continuation after this fixed prefix. Unless otherwise noted, “The image shows” is the default continuation prefix used throughout this section.

The local reference run uses `Qwen/Qwen-1.8B-chat`, a 24-layer decoder-only model. Its steering vector is learned from 3,000 COCO-derived training captions: 1,500 spatial captions and 1,500 descriptive captions. This local run is used for layer selection, continuation-prefix comparisons, next-token diagnostics, and qualitative analysis.

The standardized model comparison uses the local `Qwen/Qwen-1.8B-chat` reference run and two additional base Qwen models: `Qwen2.5 3B base` and `Qwen2.5 7B base`. For each model, a new steering vector is extracted from that model’s own activations, a steering layer is selected within that same model, and that model is then steered during generation. These runs are therefore a generalization check for the steering procedure, not a test of direct vector transfer across architectures.

The standardized model comparison is designed to match the local `Qwen/Qwen-1.8B-chat` “The image shows” experiment as closely as possible. The runs use the same prompt format, the

same default continuation prefix, the same full-sequence steering setup, the same normalized spatial ratio metric, and the same degeneration detection method. The intended differences are only the model itself and the model-specific steering vector and selected layer extracted within that model. This makes the comparison directly tied to the local “The image shows” condition rather than to a different prompt condition.

The validation data comes from a separate held-out pool. After the training captions are selected, the next highest-scoring filtered captions are used to make a validation pool of 1,500 captions: 750 spatial and 750 descriptive. For the local Qwen run, 1,000 captions are sampled from this pool. The final cached validation sample contains 486 spatial captions and 514 descriptive captions. The standardized model comparison uses this same validation sample so that differences reflect model behavior rather than different evaluation data.

The main evaluation uses full-sequence steering. In this setting, the steering intervention is applied at every generated token. For each model, continuation prefix, and steering coefficient, the generated continuation set Y is scored using the mean normalized spatial ratio $\bar{R}(Y)$. The next-token disparity score is used for prompt filtering, layer selection, and diagnostic checks, but the main output metric is computed from the full generated continuations.

I also track obvious degeneration as a coarse automatic quality screen. An output is labeled degenerate if it contains severe repetition, punctuation collapse, or malformed low-information text that no longer functions as a usable continuation. This detector is not a substitute for human evaluation of quality, faithfulness, or accessibility usefulness; it is used only to identify cases where steering has clearly broken generation.

The only place where the default continuation prefix is deliberately changed is the later local continuation-prefix analysis. In that subsection, “The image shows” is replaced with “The scene depicts” or “Positioned” in order to test how prefix choice changes steering behavior.

3.2 Standardized Qwen Model Comparison

The first experiment tests whether the steering procedure produces a similar continuation-level effect across Qwen models when the prompt format and continuation prefix are held fixed. All runs in this comparison use the default “The image shows” template. The compared models are the local Qwen/Qwen-1.8B-chat reference run, Qwen2.5 3B base, and Qwen2.5 7B base.

For each model, the steering vector is learned and applied within that same model. The comparison therefore provides a limited within-family check of whether the procedure behaves similarly across Qwen models under matched prompt conditions. It does not test whether a single steering vector transfers directly from one model to another.

Figure 2 shows the coefficient-response curves for the standardized comparison. The local Qwen/Qwen-1.8B-chat run shows the strongest response: it moves from a descriptively weighted baseline to $\bar{R} = +0.231$ at $\lambda = +20$. The larger base models respond more weakly under the same “The image shows” prefix. Qwen2.5 3B base reaches only $\bar{R} = +0.013$ at the best tested setting, while Qwen2.5 7B base remains slightly negative even at the largest tested positive coefficient.

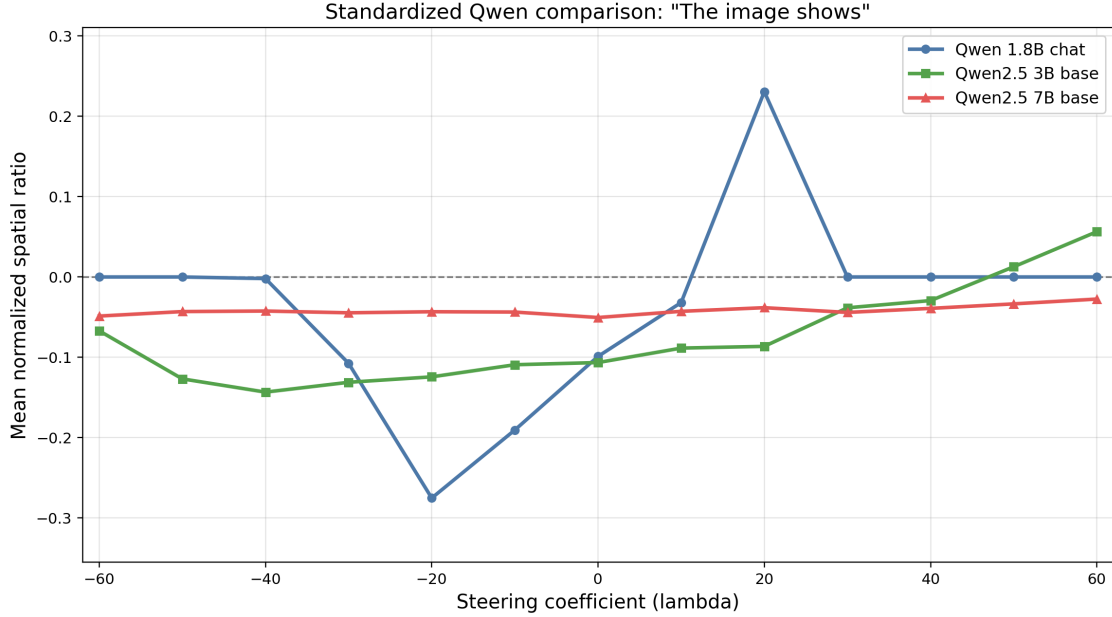


Figure 2: Standardized Qwen model comparison using the fixed continuation prefix “The image shows.” Each line shows the mean normalized spatial ratio across steering coefficients. Each model is steered using a vector separately extracted from that model’s own activations. Under this less spatially primed prefix, the local Qwen/Qwen-1_8B-chat run shows the clearest steering response, while the larger base models respond more weakly.

Model	Best λ	$\bar{R}$ at $\lambda = 0$	$\bar{R}$ at best λ
Qwen/Qwen-1_8B-chat	+20	-0.099	+0.231
Qwen2.5 3B base	+50	-0.105	+0.013
Qwen2.5 7B base	+60	-0.048	-0.028

Table 1: Best tested coefficients for the standardized Qwen comparison under the fixed “The image shows” continuation prefix. Here, “best” means the tested coefficient with the highest mean normalized spatial ratio among the evaluated coefficients. The larger base models are more stable across steering coefficients but show smaller continuation-level shifts than the local Qwen/Qwen-1_8B-chat reference run.

This comparison shows that the steering procedure does not generalize uniformly across Qwen models under the less spatially primed “The image shows” prefix. The local Qwen/Qwen-1_8B-chat reference run shows the clearest continuation-level shift, while the larger base models remain closer to their baselines across the tested coefficient range. This suggests that model choice and prompt wording both matter: larger models may be more stable under large steering coefficients, but they do not automatically show stronger spatial steering under the same prompt condition.

3.3 Local Next-Token Steering Diagnostic

Before analyzing full generated continuations under different prefixes, I use a next-token diagnostic to check whether the selected local steering direction affects the measured spatial–descriptive signal under the default prefix. Figure 3 restricts this diagnostic to the local Qwen/Qwen-1.8B-chat run with the continuation prefix “The image shows.”

The intervention coefficient $\lambda = 20$ is used because it gives the lowest RMS next-token disparity in a sweep over $\lambda \in \{10, 15, 20, 25, 30, 35, 40\}$. At this setting, RMS disparity drops from 0.79 to 0.56, a 28.9% reduction. This should be interpreted as controlled positive steering rather than exact centering: the mean disparity shifts from slightly descriptive to spatial, while the extreme saturated bands are reduced.

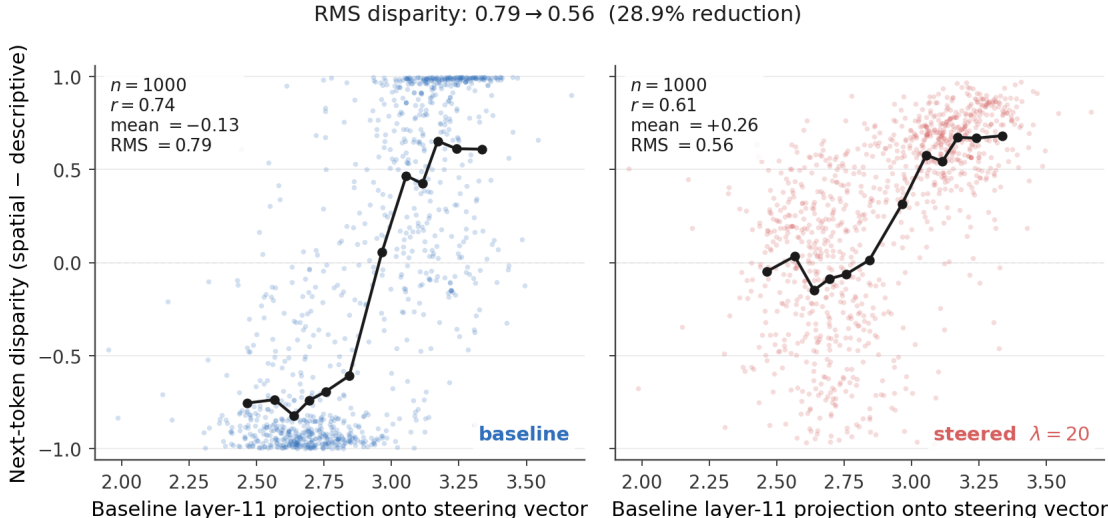


Figure 3: Before/after next-token diagnostic for the local Qwen/Qwen-1.8B-chat run, restricted to the continuation prefix “The image shows.” Each point is one held-out COCO caption from the saved 1,000-caption validation split. Both panels use the baseline layer-11 scalar projection onto the selected steering vector as the x -axis; the y -axis is the constrained next-token spatial-minus-descriptive disparity. At $\lambda = 20$, the intervention reduces RMS disparity from 0.79 to 0.56, a 28.9% reduction, while shifting the mean disparity from -0.13 to $+0.26$. The black line shows the binned mean over projection quantiles.

This diagnostic is not the main output-quality evaluation. It measures how the selected direction affects the model’s next-token distribution under one fixed continuation prefix. The following subsections evaluate complete generated continuations, where steering can improve the spatial-language score but can also damage coherence or faithfulness at larger coefficients.

3.4 Local Continuation-Prefix Response and Saturation

After the standardized model comparison, the local Qwen/Qwen-1.8B-chat run is used to examine how changing the continuation prefix affects steering behavior. Figure 4 plots the mean normalized spatial ratio across steering coefficients on the full 1,000-caption validation set. This subsection is the only place where the default “The image shows” continuation prefix is modified.

In this subsection, the same local layer-11 steering vector is used; only the continuation prefix supplied before generation is changed.

The three continuation prefixes begin from different baselines. “The image shows” and “The scene depicts” are the main test cases because they are less spatially primed. “Positioned” is included as a spatially primed diagnostic because the word itself already pushes the model toward spatial language before steering is applied. This makes it useful for comparison, but less useful as a clean test of what the steering vector adds.

At $\lambda = 0$, the two main continuation prefixes start descriptively weighted: $\bar{R} = -0.0989$ for “The image shows” and $\bar{R} = -0.1396$ for “The scene depicts”. Positive steering moves both prefixes toward spatial language. The strongest mean spatial ratios occur at $\lambda = +20$, with $\bar{R} = 0.2306$ for “The image shows” and $\bar{R} = 0.1000$ for “The scene depicts”. Negative steering has the opposite effect: both prefixes become most descriptive at $\lambda = -20$, reaching $\bar{R} = -0.2752$ and $\bar{R} = -0.2248$, respectively. “Positioned” follows the same general pattern but peaks earlier, reaching $\bar{R} = 0.1683$ at $\lambda = +10$ and its lowest value, $\bar{R} = -0.1717$, at $\lambda = -20$.

Overall, larger steering values are not always better. Moderate positive steering can shift outputs toward spatial language, but larger coefficients do not continue to improve the continuation ratio and can sharply increase degeneration. At $\lambda = +20$, degeneration is essentially absent for both main prefixes at 1/1,000, but rises sharply to 448/1,000 and 453/1,000 at $\lambda = +30$.

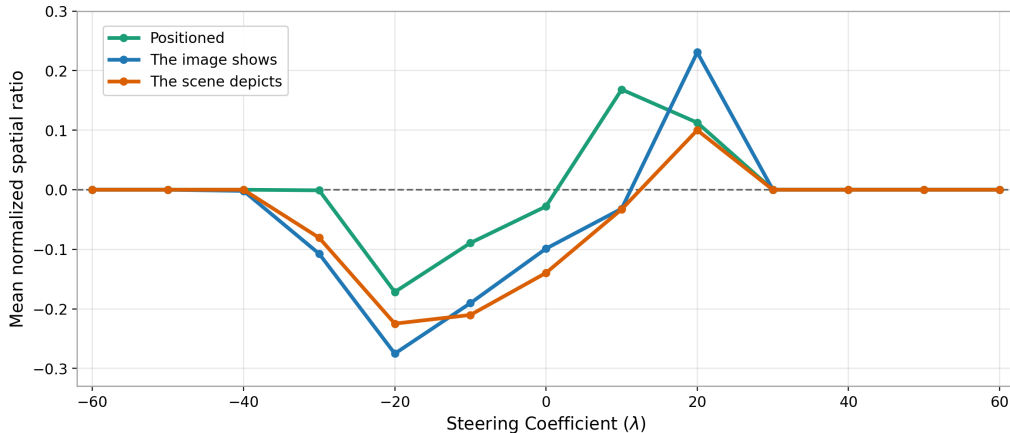


Figure 4: Coefficient-response curves on the full validation set for the local Qwen/Qwen-1.8B-chat run. Each line shows the mean normalized spatial ratio for one continuation prefix across steering coefficients. Ratios closer to +1 are more spatial, ratios closer to -1 are more descriptive, and the dashed horizontal line marks the crossover point at $\bar{R} = 0$. This figure reports the word-list ratio only; degeneration and coherence are discussed separately in the text.

3.5 What Steering Changes in the Text

Table 2 and Table 3 show controlled bidirectional examples from the local Qwen/Qwen-1.8B-chat run. In each example, the input caption and continuation prefix are held fixed, and only the steering coefficient changes. The rows are exact generated continuations, so some outputs end mid-sentence because the sweep used a fixed token budget. These examples intentionally include extreme steering

coefficients to show where the intervention stops producing useful text and begins to degenerate. **Color key:** **blue bold** terms are spatial word-list terms, and **orange bold** terms are descriptive word-list terms.

Input caption: “A white cat sits on a large red object by a river.”

Continuation prefix: “The image shows”

λ	Generated continuation
-40	“no other any other any other any other any other any other any other any other any other any other any other”
-20	“a single white cat sitting on a large red object by a river. There is no other description mentioned”
0	“a white cat sitting on a large red object by a river. The cat’s fur is white ,”
+10	“a serene and peaceful scene, with the white cat perched on top of a large red object that”
+20	“a serene and tranquil scene, with the cat sitting on the right side of the picture, against the”
+30	“, the viewer is of the, how the, the, the, the, the, the,”

Table 2: Controlled bidirectional steering example with fixed input caption and continuation prefix. Spatial and descriptive word-list terms are highlighted.

In this example, negative steering keeps the continuation close to appearance and object identity: the continuation stays close to the caption by describing the **white** cat and **red** object. The baseline is also faithful but remains mostly appearance-focused. Moderate positive steering introduces spatial layout language, such as the cat being on **top** of the object and later on the **right side** of the picture. At stronger coefficients, the output becomes generic and unfinished, and by $\lambda = +30$ it degenerates into repeated function-word fragments.

Input caption: “Motorcycle parked in front of a tent with the sun going down behind them.”

Continuation prefix: “The image shows”

λ	Generated continuation
-40	“no other any other any other any other any other any other any other any other any other any other any other any other”
-20	“a motorcycle parked in front of a tent and the sun going down behind it. There is no other”
0	“a motorcycle parked in front of a tent, which suggests that it is likely a camping or outdoor recreational”
+10	“a motorcycle parked in front of a tent, with the sun setting in the background . The scene is”
+20	“a scene of a motorcycle parked in front of a tent, with the sun setting in the back-ground .”
+30	“, captured by the canvas, is, we see,, the aftermath of a, of a,”
+40	“, is,,,,,,,,,,,,,”

Table 3: Controlled bidirectional steering example with fixed input caption and continuation prefix. Spatial and descriptive word-list terms are highlighted.

This example shows a different pattern. Moderate negative steering and the baseline remain mostly literal, while moderate positive steering preserves and foregrounds the background relation between the motorcycle, tent, and setting sun. However, the useful range is limited: by $\lambda = +30$, the continuation becomes meta and low-information, and by $\lambda = +40$, it collapses into punctuation. This supports the broader result that stronger steering can damage output quality even when moderate steering adds spatial detail.

4 Discussion

The experiments support four main findings. First, the local Qwen/Qwen-1.8B-chat reference run contains a measurable spatial-descriptive signal in the middle layers. Layers 11–13 give the strongest validation signal, which suggests that the measured spatial-descriptive language axis is not only visible in the output distribution but also recoverable from the model’s residual-stream activations.

Second, the selected steering direction can shift generated continuations toward spatial language without changing model weights. The strongest local continuation-level result comes from the “The image shows” continuation prefix, which shifts from $\bar{R} = -0.0989$ at baseline to $\bar{R} = +0.2306$ at $\lambda = +20$, with only 1 out of 1,000 outputs labeled degenerate. The next-token diagnostic shows a related effect: at $\lambda = 20$, RMS next-token disparity drops from 0.79 to 0.56. These results support the main claim that activation steering can partly shift a measured spatial-descriptive language signal in caption-conditioned generation.

Third, larger steering coefficients are not always better. Moderate positive steering can add useful layout information, as shown in the controlled qualitative examples. However, stronger steering can make outputs more generic, unfinished, repetitive, or less faithful to the input caption. This also appears in the quantitative results: at $\lambda = +20$, degeneration is nearly absent for the two main continuation prefixes, but at $\lambda = +30$, degeneration rises sharply. A high spatial ratio is therefore only meaningful when the generated text remains coherent and connected to the prompt.

Fourth, prompt wording affects how the results should be interpreted. The “The image shows” and “The scene depicts” prefixes are the clearest local tests because they are less spatially primed at baseline. By contrast, “Positioned” already encourages spatial language before steering is applied, so results under that prefix are useful as a diagnostic but less clean as evidence of what the steering vector adds by itself.

The standardized Qwen model comparison provides a generalization check for the procedure under a fixed prompt condition. In this comparison, each model is steered using a vector separately extracted from that model’s own activations; no vector is transferred across models. Under the shared “The image shows” prefix, the local Qwen/Qwen-1.8B-chat run shows the strongest continuation-level shift, while Qwen2.5 3B base and Qwen2.5 7B base respond more weakly. This suggests that the procedure is model-dependent: matching the prompt template and metric is not enough to guarantee that larger base models will show stronger spatial steering.

From an accessibility perspective, the results are encouraging but incomplete. The experiments show that spatial language is partly steerable at inference time in a caption-continuation setting, which could be relevant for future systems that aim to provide more layout information to blind and low-vision users. However, counting more spatial word-list terms does not automatically mean a description is more useful. The results should therefore be read as evidence that a relevant model behavior can be measured and steered, not as proof that the resulting descriptions are already accessibility-ready.

5 Limitations

The first limitation is prompt dependence. The local results vary across continuation prefixes, and the “Positioned” prefix is already spatially primed before steering is applied. This makes large spatial ratios under that prefix harder to interpret as evidence of steering alone. The more reliable local evidence comes from “The image shows” and “The scene depicts”, where the prefix itself is less spatially loaded.

The second limitation is the metric. The normalized spatial ratio counts terms from the spatial and descriptive word lists in the generated continuation. This is useful for measuring directional change, but it is still only a proxy. A continuation can score higher on this metric without becoming more useful to a blind or low-vision user. The metric also depends on single-token word-list matching, while many spatial expressions are multi-word phrases, such as “in front of”, “next to”, “far away”, and “close by”. Depending on tokenization and wording, these phrases may be missed or only partially counted.

The degeneration detector is also limited. It is useful for identifying obvious failures such as severe repetition, punctuation collapse, or malformed continuations, but it does not measure factual accuracy, faithfulness to the input caption, readability, or accessibility usefulness. Human evaluation would be needed to judge whether a steered continuation is actually better for a user.

The third limitation is dataset construction. The caption pools are built using lexical filters and hand-curated word lists. This makes the spatial-descriptive contrast easier to study, but it also means the learned direction depends on the vocabulary choices made in this project. Future work could improve this by using a semantic retriever, a learned classifier, or a richer phrase-level inventory that better captures spatial relations beyond individual words.

The fourth limitation is the text-only evaluation setup. The model is given COCO captions, not the

original images. This means the experiment tests caption-conditioned continuation rather than full image understanding. The accessibility motivation concerns instructional and layout-relevant image descriptions, but the experiments do not evaluate a real image-description system on instructional images. Future work should test the method with actual image inputs or with datasets designed around task-relevant instructional scenes.

The fifth limitation is the scope of the standardized Qwen model comparison. These runs do not test whether one vector transfers across architectures. Each model uses its own separately extracted steering vector, and the comparison is limited to Qwen-family models under one fixed continuation prefix, “The image shows.” The larger base models show weaker continuation-level shifts than the local reference model, but this does not prove that they are inherently less steerable. A wider coefficient range, additional prompt templates, or separate tuning of model-specific layer choices could change the result.

Finally, this paper is motivated by accessibility for blind and low-vision users, but it does not include user studies, human preference judgments, or task-based evaluation with blind participants. The experiments show that spatial language is partly controllable along a measured language axis. They do not prove that the resulting descriptions are more useful in practice. Demonstrating that steered descriptions actually help blind or low-vision users understand scene structure remains the most important next step.

6 Conclusion

This paper asks a narrow question: can activation steering identify and control a spatial-versus-descriptive direction in caption-conditioned generation? The answer is yes, but with important caveats.

The local Qwen/Qwen-1.8B-chat reference run shows a clear spatial-descriptive signal in the middle layers. The full 1,000-caption validation sweep shows that moderate steering can shift average outputs from descriptively weighted to spatially weighted with almost no degeneration. The strongest local result comes from the “The image shows” continuation prefix, which moves from $\bar{R} = -0.0989$ at baseline to $\bar{R} = +0.2306$ at $\lambda = +20$, with only 1 out of 1,000 outputs labeled degenerate. The controlled qualitative examples show the same pattern in individual cases: moderate steering can add layout details, while stronger steering can become more generic or less faithful.

The standardized Qwen model comparison gives a more cautious result. Under the fixed “The image shows” prefix, the local Qwen/Qwen-1.8B-chat run shows the clearest steering response, while Qwen2.5 3B base and Qwen2.5 7B base respond more weakly. Because each model uses a separately extracted steering vector, these results should not be interpreted as direct transfer of one residual-stream vector across architectures. Instead, they show that the procedure is sensitive to model choice as well as prompt wording.

The main claim is therefore not that the system already produces accessibility-ready image descriptions. A more careful claim is that spatial language in caption continuation is measurable and partly steerable at inference time. The effect depends on prompt wording, model choice, coefficient range, and output-quality constraints. The most credible results are modest, but they show that steering can move generated language toward object locations and scene layout without retraining the model.

Three next steps follow from these results. First, future work should improve the metric by handling multi-word spatial expressions and measuring scene-level structure more directly. Second, the method should be tested on actual image-description systems and on instructional image settings where spatial relationships are task-relevant. Third, and most importantly, future work should test whether steered descriptions actually help blind or low-vision users understand scene structure. That user-centered evaluation is the accessibility motivation behind this project and remains the most important next step.

A Spatial and Descriptive Word Lists

The paper uses two word sets to define the spatial-versus-descriptive axis used in the normalized spatial ratio metric. The spatial set contains words related to position, direction, and scene layout. The descriptive set contains words related to color, size, shape, and visual appearance.

Spatial words. left, right, above, below, beside, behind, between, near, far, north, south, east, west, up, down, top, bottom, side, center, middle, adjacent, opposite, across, along, around, underneath, beneath, overhead, inside, outside, interior, exterior, within, surrounding, foreground, background, horizon, vertical, horizontal, diagonal, parallel, perpendicular, distant, close.

Descriptive words. red, blue, green, yellow, orange, purple, pink, brown, black, white, gray, violet, gold, silver, bronze, copper, beige, turquoise, large, small, tall, short, thick, thin, wide, narrow, long, tiny, huge, massive, miniature, round, square, triangular, rectangular, circular, oval, cylindrical, spherical, flat, curved, straight, bent, angular, symmetric, asymmetric, hollow, solid, bright, dark, colorful, plain, detailed, simple, clean, dirty, messy, pristine, broken, cracked.

B Code Availability

Code, plotting scripts, and experiment utilities for this project are available at: <https://github.com/bthaas/vision-bias-steering>.

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